Kona Naga Surya Siva Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
Sasi Junior College	Class XII (Board of Intermediate Education Andhra Pradesh)	2021	98.30%
Sasi (EM) High School	Class X (Board of Secondary Education Andhra Pradesh)	2019	10.00%

INTERNSHIPS

SPARK Fellowship, IIT Roorkee | Undergraduate Researcher

- Researcher in Sustainable Catalysis Lab at **IIT Roorkee** under the guidance of Dr.Niwesh Ojha.
- Synthesised **photocatalysts** for the production of **hydrogen peroxide** using water and sunlight.
- Developed calibration curves using **UV-vis spectrophotometry** and Origin software to quantify hydrogen peroxide in samples.

PROJECTS

Engineering gel polymer electrolyte for K-CO2 batteries

M.Tech Thesis Project

- Researcher in **CARBON Lab** at IIT Hyderabad under guidance of **Dr.Chandra Shekhar Sharma**.
- Engineered **gel polymer electrolytes** via **electrospinning** to address electrolyte stability challenges.
- Characterised ionic/electrochemical properties using **EIS**, **LSV**, and **K|GPE|K plating-stripping**.
- Achieved **800+ h** stable with **0-5.8 V** electrochemical stability window and **8.9 mS/cm** ionic conductivity, demonstrating enhanced electrochemical stability and ion transport.

Photoelectrochemical CO Reduction

B.Tech Thesis Project

- Designed a 5 cm² PEC flow cell (serpentine flow fields, dual-layer GDE, Nafion membrane) for CO reduction.
- Synthesized Ag-WO, ZnO-SnO and Sn-BiVO photocatalysts and benchmarked them using LSV, chronoamperometry and GC analysis.
- Achieved a near-syngas CO:H ratio (1:1.16) using a ZnO-SnO heterojunction at -1.5 V vs. Ag/AgCl.
- Identified light absorption, charge recombination and HER competition as key bottlenecks.

Acetic Anhydride Plant Design

- Simulated a commercial-scale acetic anhydride production plant using the selected Eastman/Halcon process in Aspen Plus.
- Optimized key process and unit-operation parameters, achieving a product purity of >99.5%.
- Conducted process economic analysis, demonstrating strong economic viability with a 3.17-year payback period.

Computational Modeling of Incompressible Channel Flow

- Developed a **Python-based 2D CFD solver** using the **vorticity-stream function** formulation for incompressible channel flow.
- Implemented finite-difference schemes, **vectorised NumPy** computations, Gauss-Seidel iteration, and time-stepping algorithms for numerical flow simulation.
- Performed transient flow analysis and visualized velocity profiles using **NumPy** and **Matplotlib**.

Machine Learning-Based Heat Transfer Prediction of Nanofluids

- Developed an **Artificial Neural Network (ANN)** in Python to predict heat-transfer coefficients of nanofluids.
- Compared Adam and RMSProp optimisers and evaluated key **ANN parameters** for improved prediction performance.
- Applied **K-means clustering** to find the optimum number of clusters using the **elbow method**.

SKILLS

- **Programming & Computational Skills:** C, C++, Python, MATLAB, Data Structures & Algorithms, Machine Learning.
- **Chemical Engineering Software:** Aspen Plus, ANSYS Fluent
- **Soft Skills:** Leadership, Management, Teamwork, Problem-Solving, Logical & Creative Thinking, Communication

RELEVANT COURSES

- **Computing & Data:** Programming and Data Structures, Introduction to Data Science, Machine Learning for Process Systems Engineering
- **Modeling & Simulation:** Advanced Numerical Methods, Computational Fluid Dynamics, Advanced Transport Phenomena, Molecular Thermodynamics
- **Reaction & Process Engineering:** Heterogeneous Catalytic Reaction Engineering, Principles of Heterogeneous Catalysis, Advanced Process Control, Process Integration, Sustainability in Chemical Engineering

PUBLICATIONS

SSS Ganti, PK Roy, Nayonay Wagh, ***Kona Naga Surya Siva Sai*** *and Sushant Kumar* : "Strategically

Designed Catalysts for Ammonia Synthesis Under Mild Conditions: Recent Advances and

Challenges", Journal of Materials Chemistry A (Impact Factor- 10.7) -Link

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) for Applied Mathematics in Chemical Engineering, Fluid Mechanics, and Process Engineering Lab.

Volunteer at NSS (National Service Scheme) IITP- Took part in cleanliness drives to create awareness about sanitation and hygiene.

HOBBIES & INTERESTS

Multiplayer mobile gaming, Playing cricket, Listening songs, Solitude walking